

Agenda :

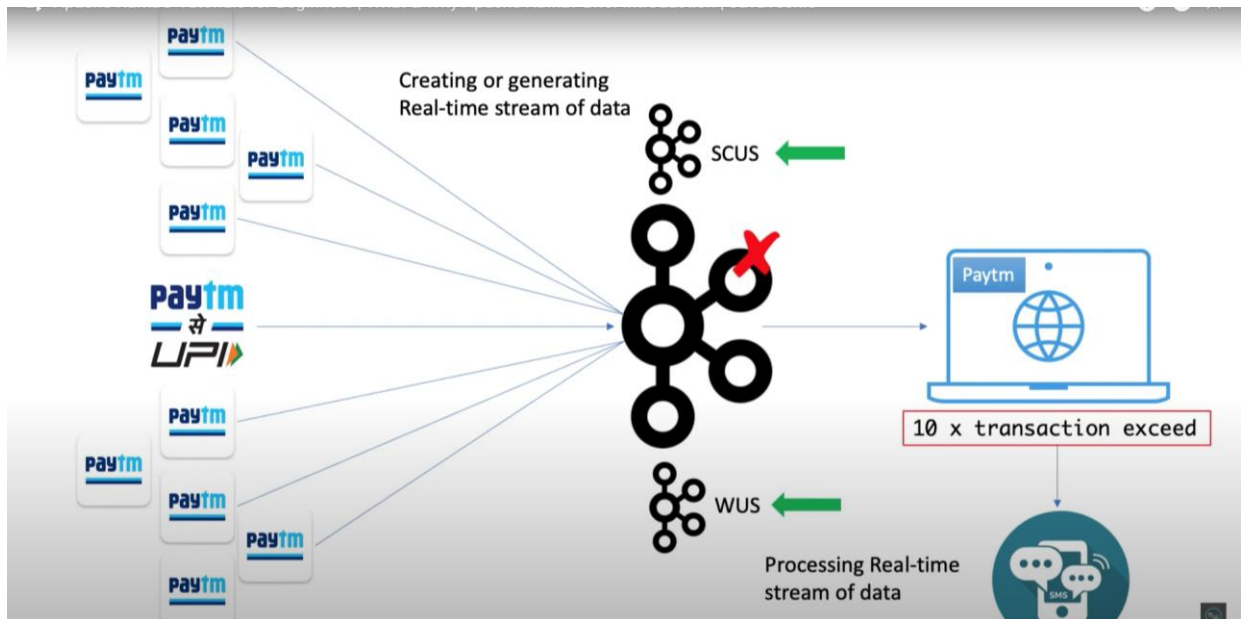
- What is Kafka ?
- Where does Kafka come from ?
- Why do we need Kafka ?
- How does it work (High-level overview)

What is Kafka ?

- Apache Kafka is an open-source distributed event streaming platform

Creating Real-time Stream

Processing Real-time Stream

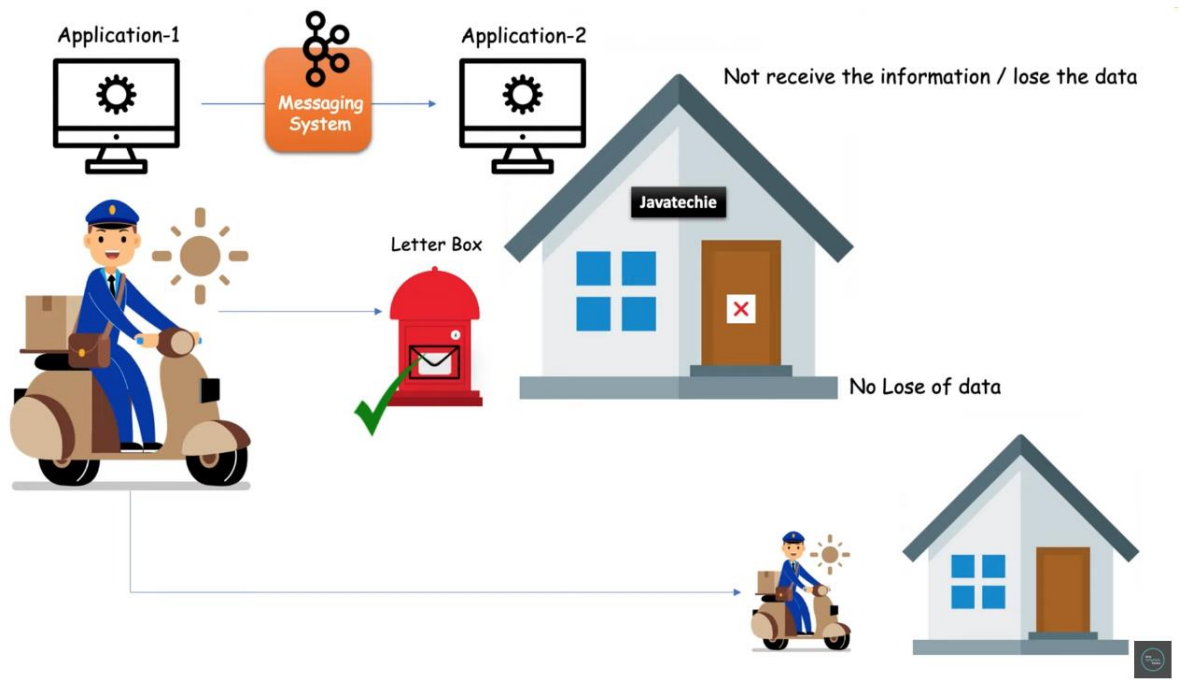


Where does Kafka come from ?

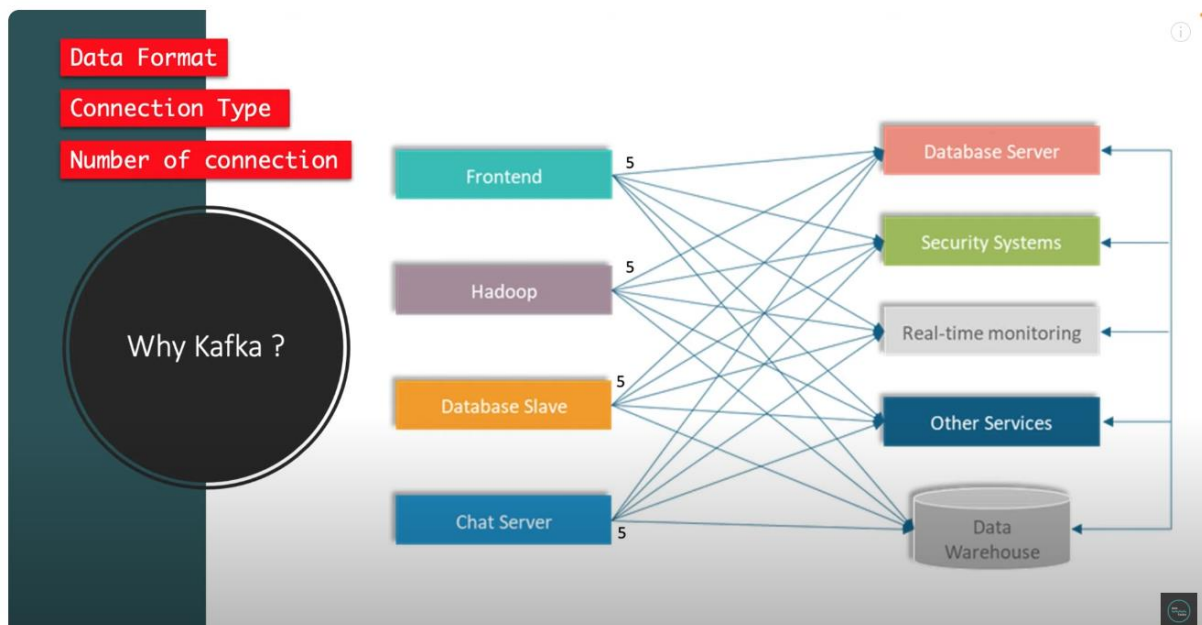
Kafka was originally developed at [in](#), and was subsequently open sourced in early **2011**



Why do we need Kafka

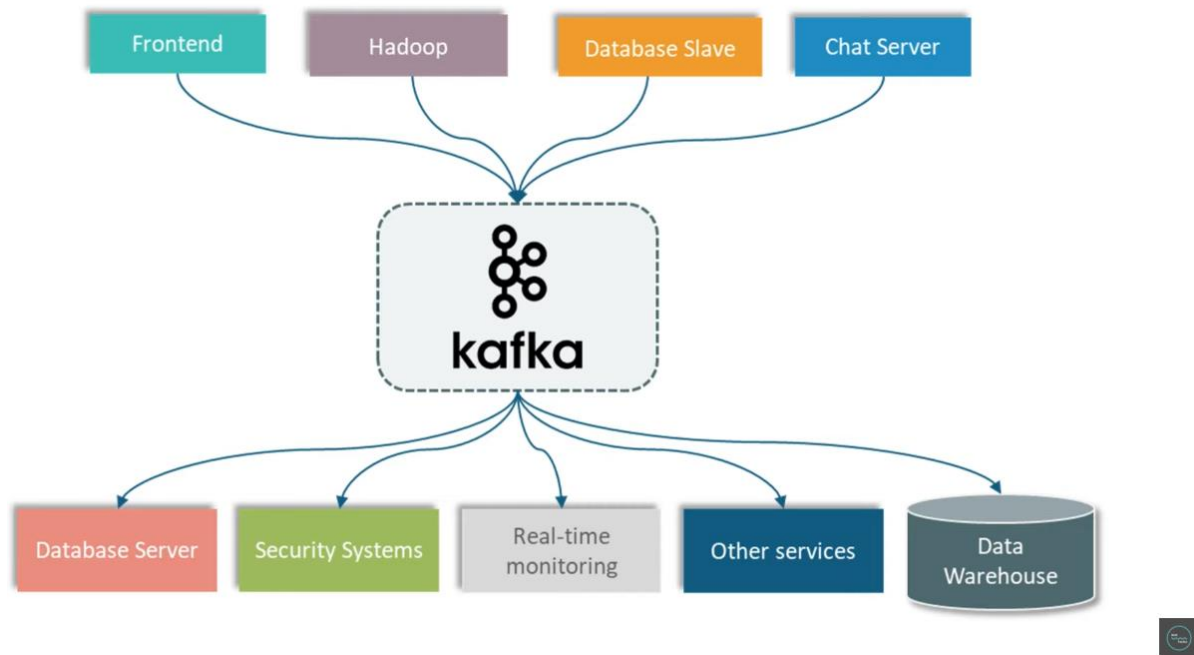


Why Kafka



In above diagram we need 20 connections to communicate with each application to reduce those bottle neck production issues/problems we need to go for solution for that we need kafka, Redis, RabbitMQ etc.,

Distributed messaging system.



Kafka

<https://kafka.apache.org/>

(download latest tar file from the website)

The screenshot shows the Apache Kafka download page in a web browser. The browser's address bar displays 'kafka.apache.org/downloads'. The page features the Kafka logo and navigation links: 'GET STARTED', 'DOCS', 'POWERED BY', 'COMMUNITY', and 'APACHE'. A yellow box highlights the 'DOWNLOAD KAFKA' button. The main heading is 'DOWNLOAD', followed by a paragraph explaining the release cycle and verification procedures. Below this is the 'SUPPORTED RELEASES' section, which lists the details for version 3.8.0, including its release date, release notes, Docker images, source download, and binary downloads. The binary downloads section lists two options: 'Scala 2.12 - kafka_2.12-3.8.0.tgz (asc, sha512)' and 'Scala 2.13 - kafka_2.13-3.8.0.tgz (asc, sha512)'. A footer note mentions building for multiple versions of Scala.

DOWNLOAD

The project goal is to have 3 releases a year, which means a release every 4 months. Bugfix releases are made as needed for supported releases only. It is possible to verify every download by following these [procedures](#) and using these [KEYS](#).

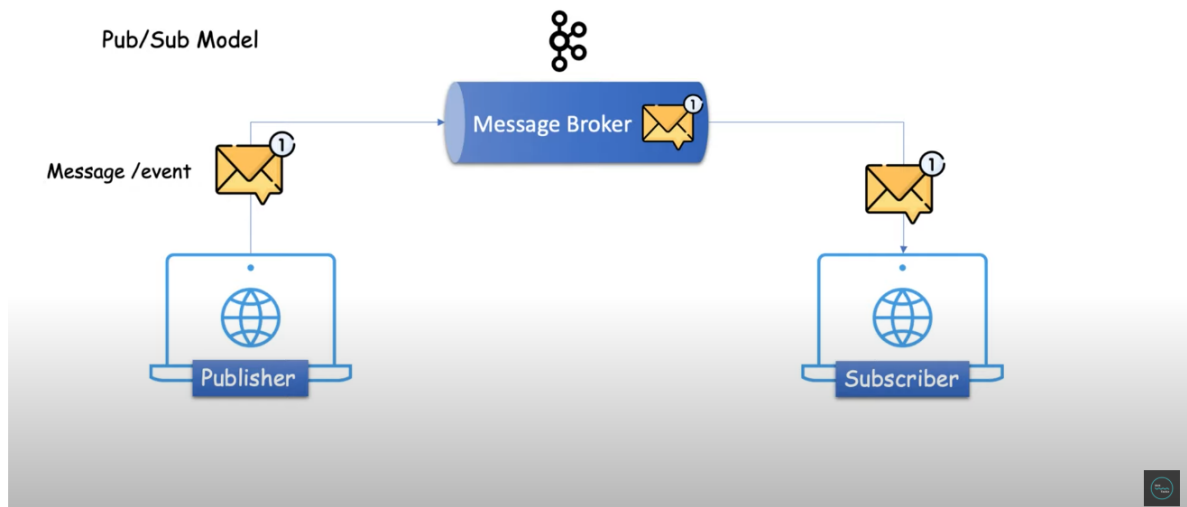
SUPPORTED RELEASES

3.8.0

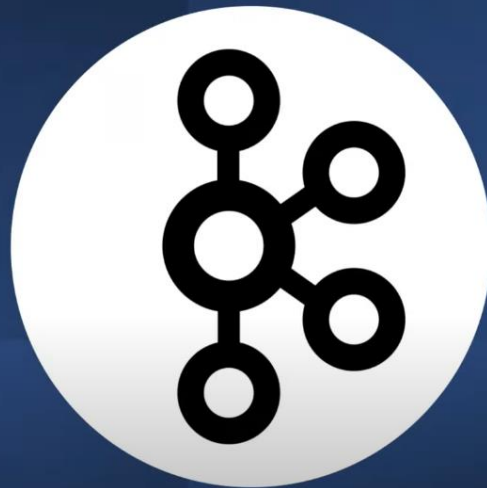
- Released July 29, 2024
- [Release Notes](#)
- Docker Image: [apache/kafka:3.8.0](#)
- Docker Native Image: [apache/kafka-native:3.8.0](#)
- Source download: [kafka-3.8.0-src.tgz \(asc, sha512\)](#)
- Binary downloads:
 - Scala 2.12 - [kafka_2.12-3.8.0.tgz \(asc, sha512\)](#)
 - Scala 2.13 - [kafka_2.13-3.8.0.tgz \(asc, sha512\)](#)

We build for multiple versions of Scala. This only matters if you are using Scala and you want a version built for the same Scala version you use.

How does it work (High-level overview)

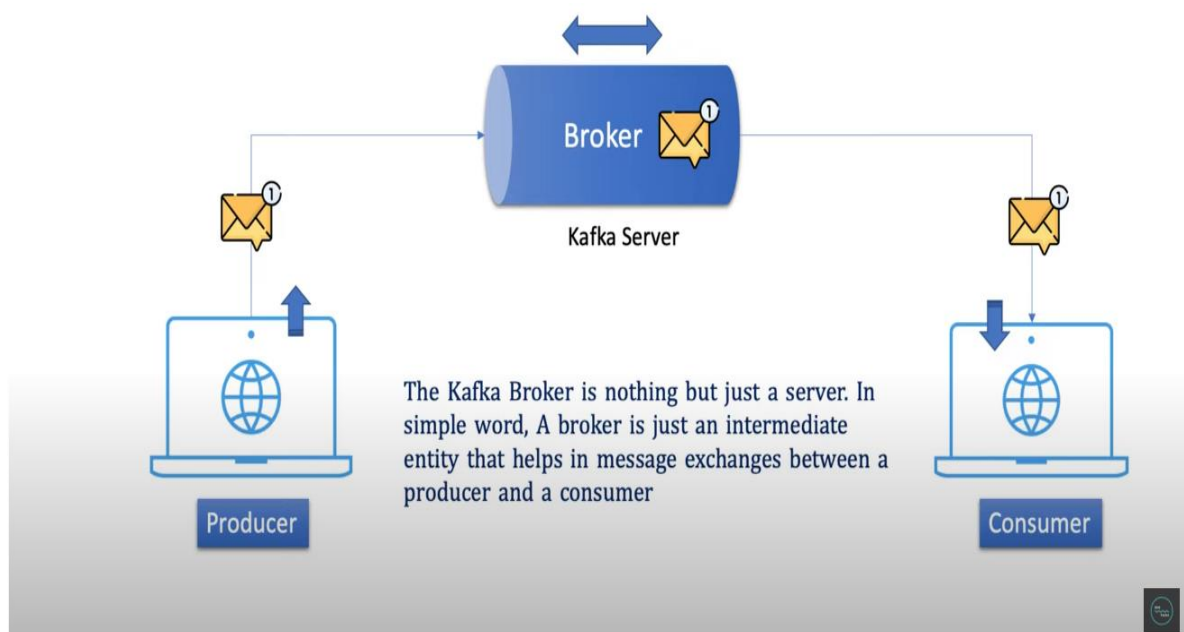


Kafka Architecture & Components

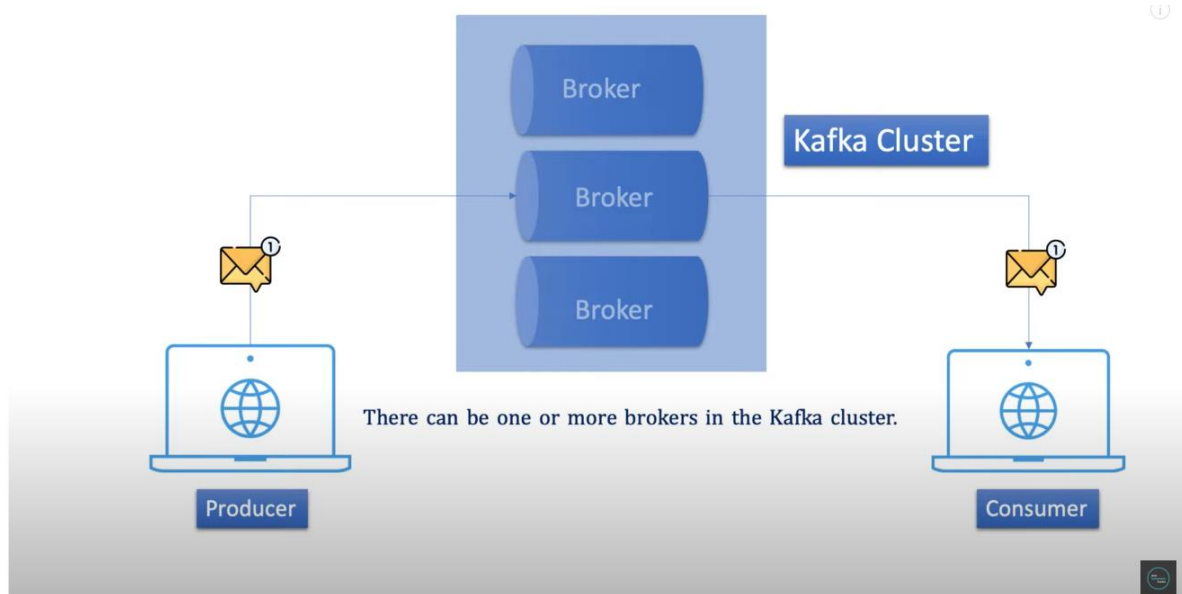


Kafka Components

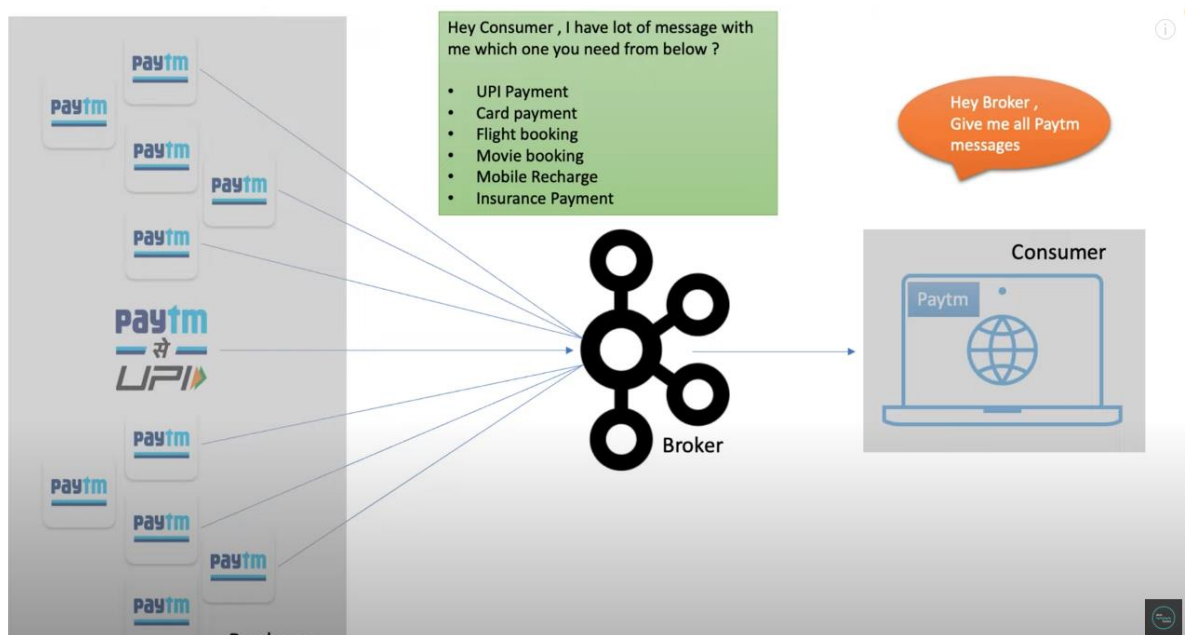
- Producer
- Consumer
- Broker
- Cluster
- Topic
- Partitions
- Offset
- Consumer Groups
- Zookeeper

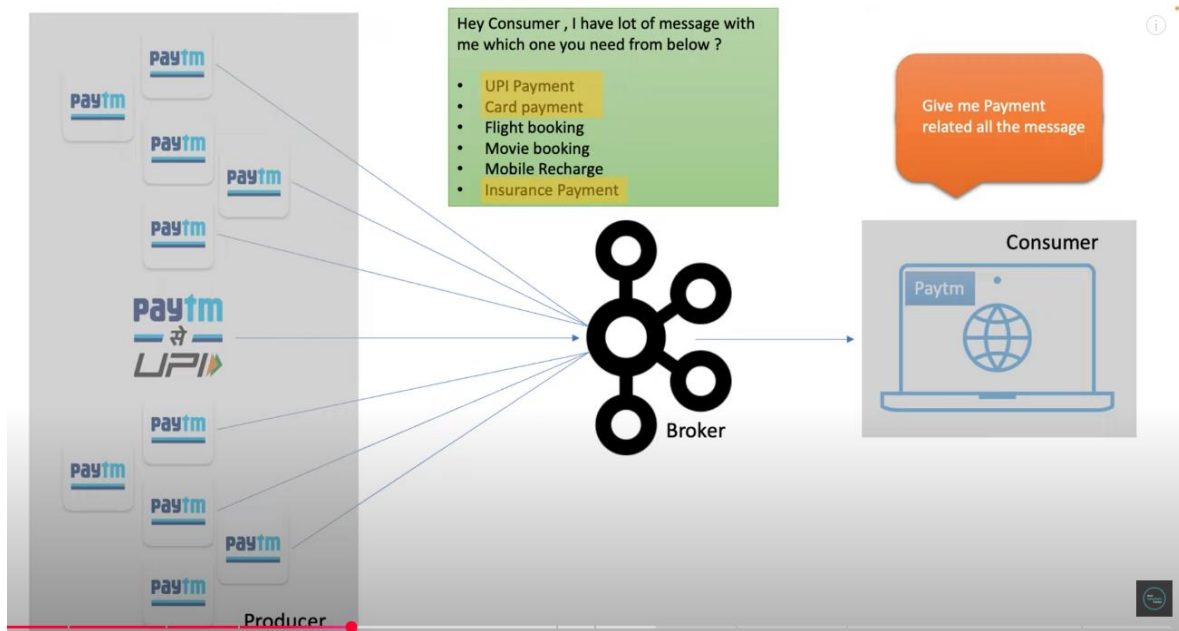


Cluster



Topic



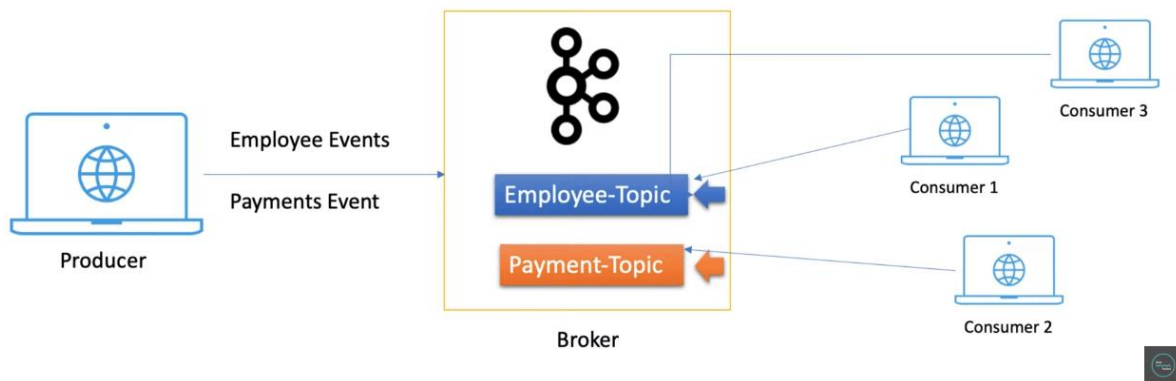


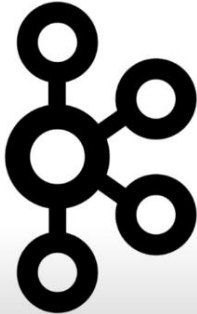
EMPLOYEE_TABLE

ID	NAME	DEPT	SALARY
1	john	IT	50000
2	sam	Bank	80000
3	joe	Admin	20000

PAYMENT_TABLE

PAYMENT_ID	CNAME	PAY MODE	SRC AC	DEST AC	AMOUNT
df364bdjfb	Basant	CREDIT	HDFC12	SBI239837	100000
bcsr72386d	Santosh	DEBIT	ICICI893	HDFC3626	890012
dfv834tjsd8	Krishna	CHEQUE	SBI2367	IDBI23726	23273582



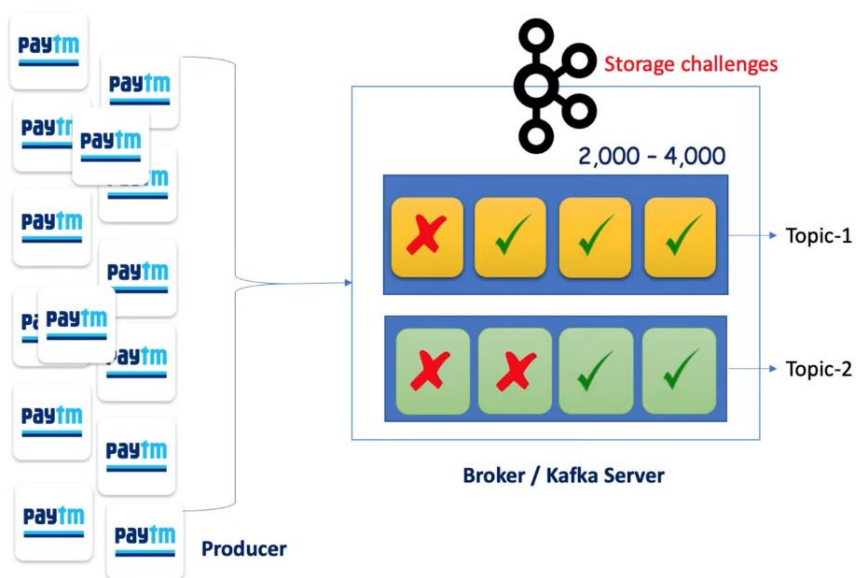


TOPIC

It specifies the category of the message or the classification of the message. Listeners can then just respond to the messages that belong to the topics they are listening on.

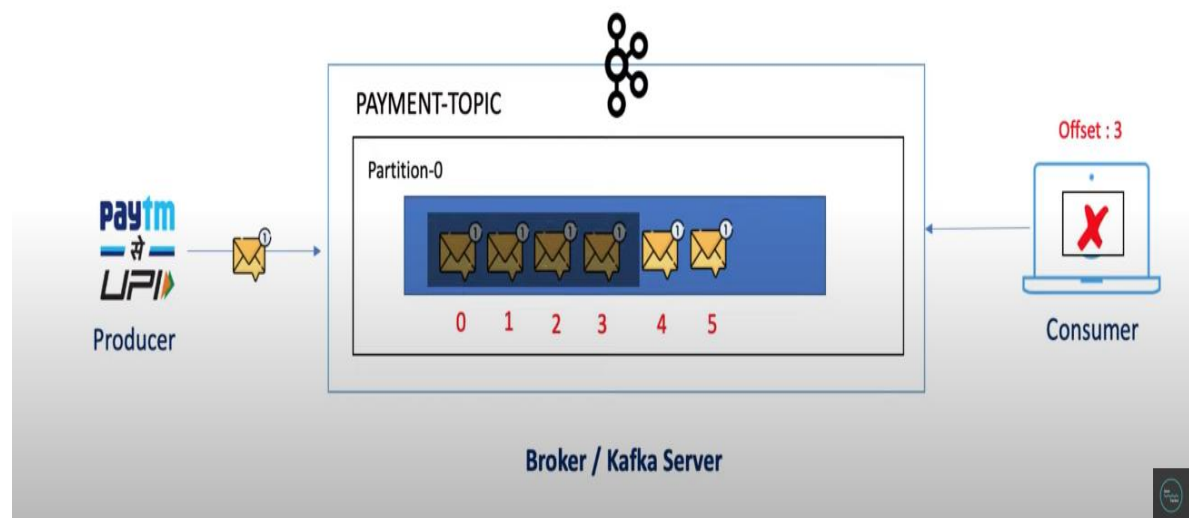
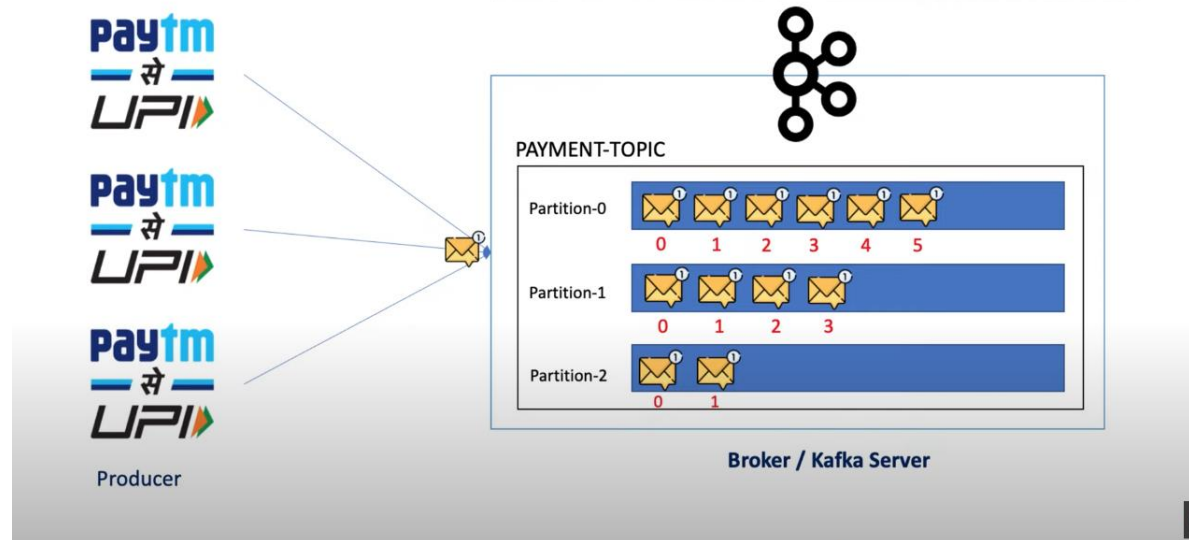
Partition

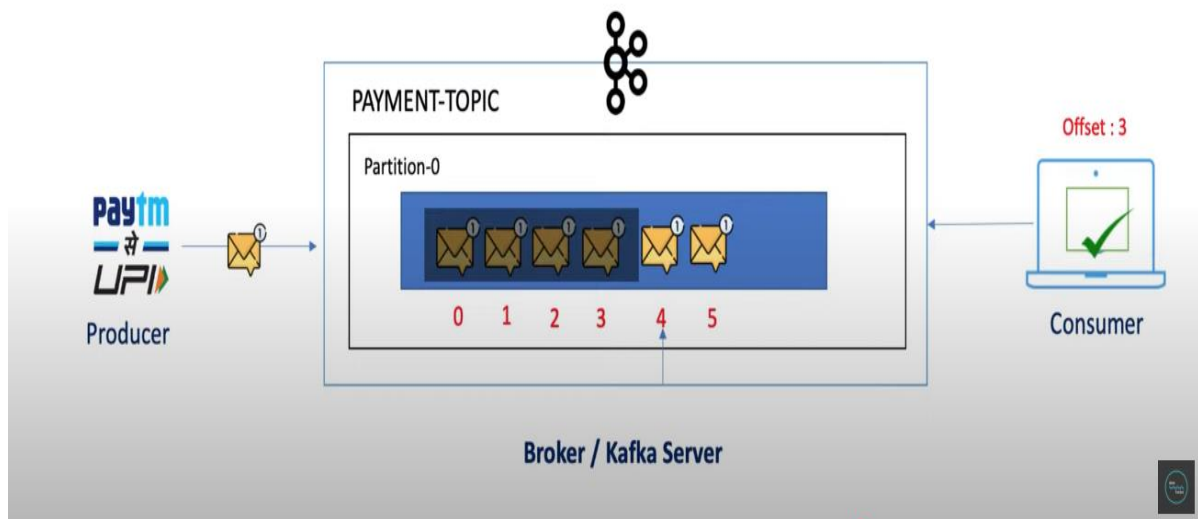
Partitions



Offset

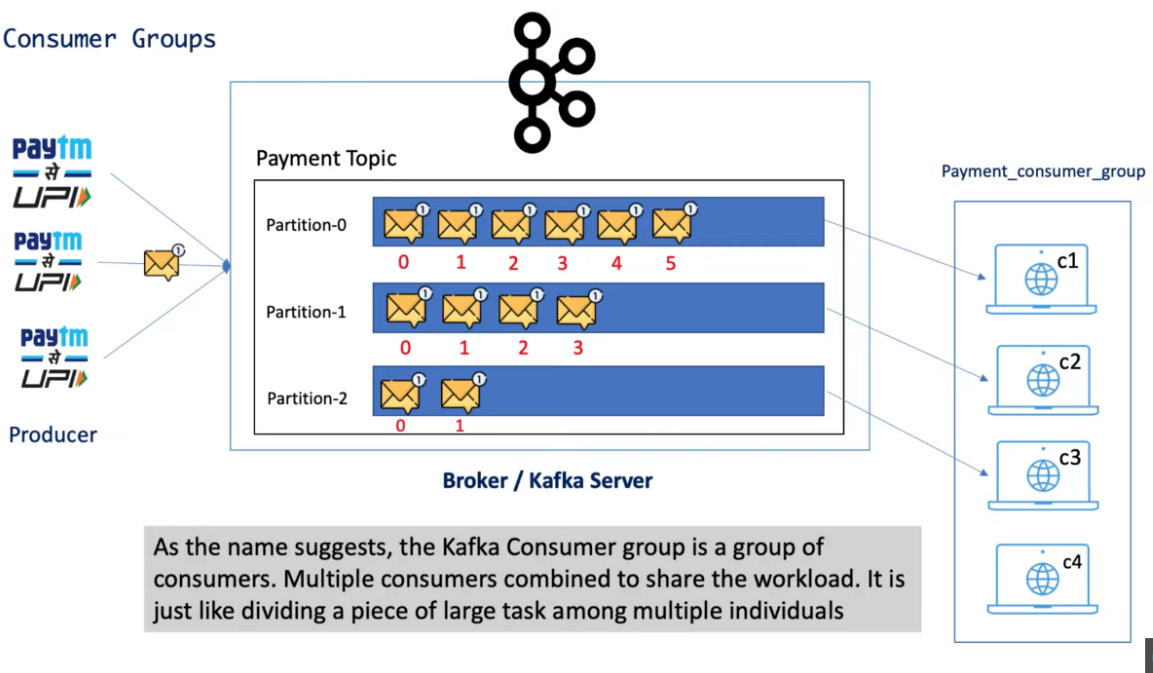
In Kafka, a sequence number is assigned to each message in each partition of a Kafka topic. This sequence number is called Offset.





Consumer Groups

Consumer Groups

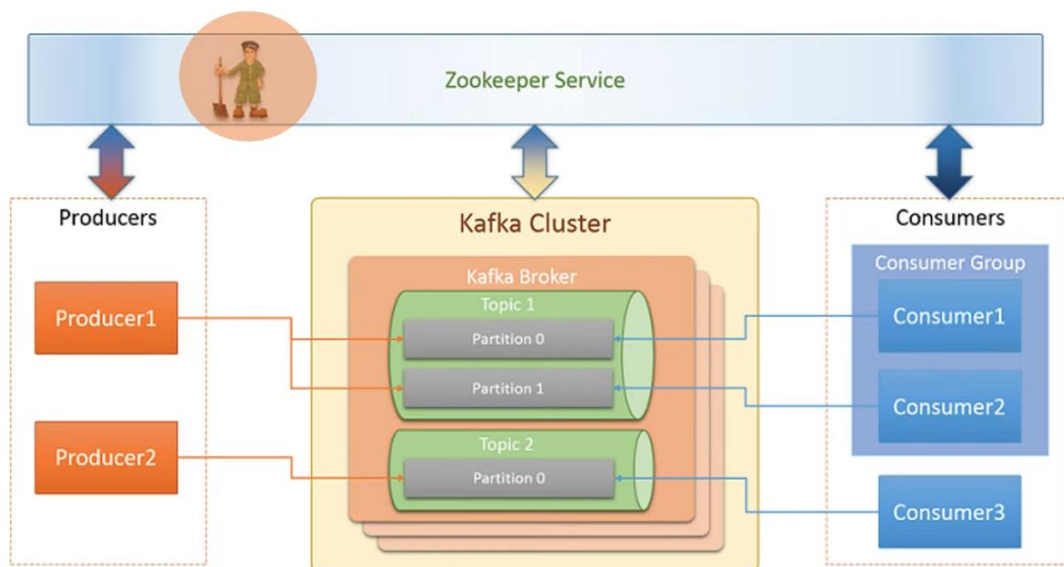


Zookeeper

Kafka Components

- Producer
- Consumer
- Broker
- Cluster
- Topic
- Partitions
- Offset
- Consumer Groups
- Zookeeper

Zookeeper is a prerequisite for Kafka. Kafka is a distributed system, and it uses Zookeeper for coordination and to track the status of Kafka cluster nodes. It also keeps track of Kafka topics, partitions, offsets, etc.



Apache Kafka Installation

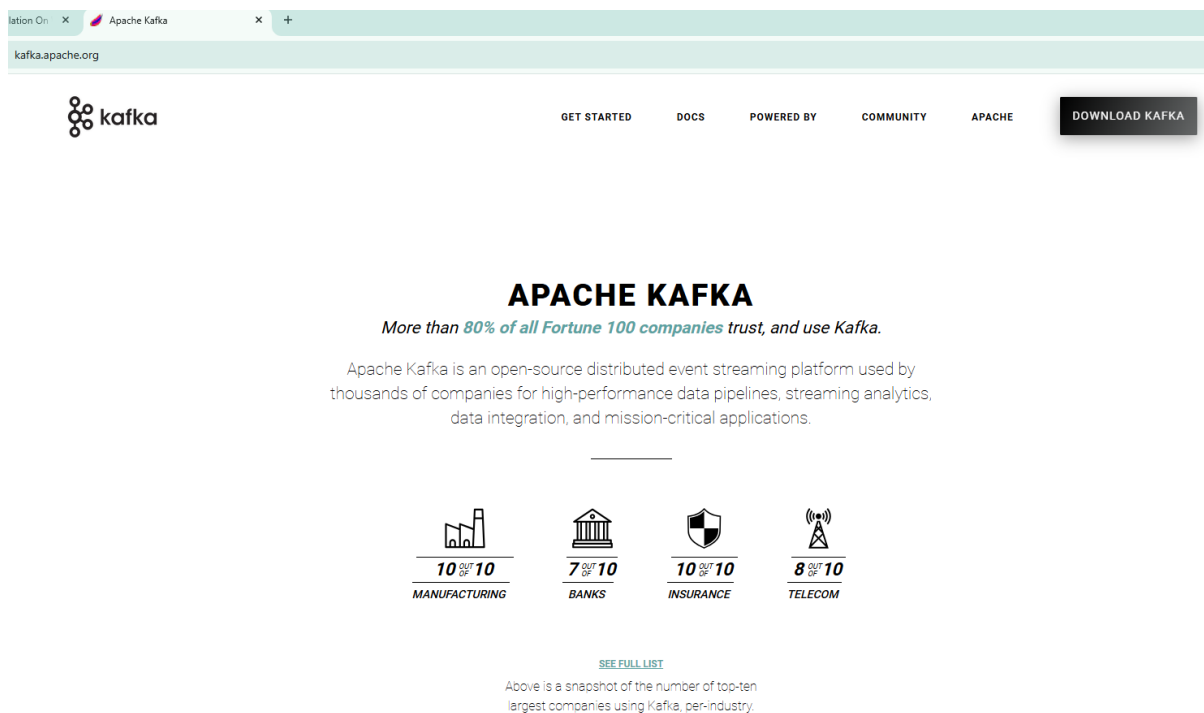
Kafka Installation

Open Source : Apache Kafka

Commercial distribution : Confluent Kafka

Managed Kafka service : confluent & AWS

<https://kafka.apache.org/>



The screenshot shows the Apache Kafka website. At the top, there's a navigation bar with links: GET STARTED, DOCS, POWERED BY, COMMUNITY, and APACHE. A 'DOWNLOAD KAFKA' button is on the right. The main heading is 'APACHE KAFKA' with a subtext: 'More than 80% of all Fortune 100 companies trust, and use Kafka.' Below this, a paragraph describes Kafka as an open-source distributed event streaming platform. A section titled '10 OUT OF 10' lists industries: MANUFACTURING, BANKS, INSURANCE, and TELECOM, each with an icon and a '10 OUT OF 10' rating. A link 'SEE FULL LIST' is provided. Below the link, a note states: 'Above is a snapshot of the number of top-ten largest companies using Kafka, per-industry.'

Industry	Rating
MANUFACTURING	10 OUT OF 10
BANKS	7 OUT OF 10
INSURANCE	10 OUT OF 10
TELECOM	8 OUT OF 10

2nd option

Confluent Apache Kafka

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DEVELOPERS
GET STARTED FREE

Local

Install Confluent Platform on a single local machine by using ZIP or TAR archives or Docker images.

ZIP	Quick Start Installation Guide	Download
TAR	Quick Start Installation Guide	Download
Docker	Installation Guide Browse Images	Quick Start

Distributed

Deploy Confluent Platform in a distributed architecture using Confluent for Kubernetes or Ansible Playbooks.

Resources to help you get started

Join Confluent Community
Interact directly with the engineers who wrote the code you just downloaded
[Join](#)

Try Confluent Cloud
Need a fully-managed solution instead?
[Try Free](#)

- Previous Versions
- Looking for RPM?
- Looking for DEB?

kafka offset explorer download

<https://www.kafkatool.com/download.html>

Offset Explorer

The Ultimate UI Tool for Kafka

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LINKS

Download

To download the Offset Explorer (formerly Kafka Tool) for your operating system, use the links below. All versions of Offset Explorer come with a bundled JRE with the exception of the Linux version. For Linux, you must have Java 11 installed on your operating system before using Offset Explorer. After downloading, refer to the [Documentation](#) to configure Offset Explorer correctly. View [change history](#) to see recent updates to Offset Explorer.

Before clicking on the download button below, carefully read the terms of the [license agreement](#). By clicking and downloading the software you are consenting to be bound by the license agreement. Do not click on download unless you agree to all terms of the [license agreement](#).

Offset Explorer is **free for personal use** only.

Offset Explorer supports Apache Kafka [®] version 0.11 and above.

Offset Explorer 3.0.2

Platform	Size	
Windows 32-Bit	81 MB	Download
Windows 64-Bit	87 MB	Download
macOS (Universal Binary)	92 MB	Download
Linux	37 MB	Download

UI Tool for Apache Kafka

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Apache Kafka is a trademark of the Apache Software Foundation. Offset Explorer is not produced by or affiliated with the Apache Software Foundation.

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Documentation

Download the tar and extract it and keep it in Some folder like C or D Drive.

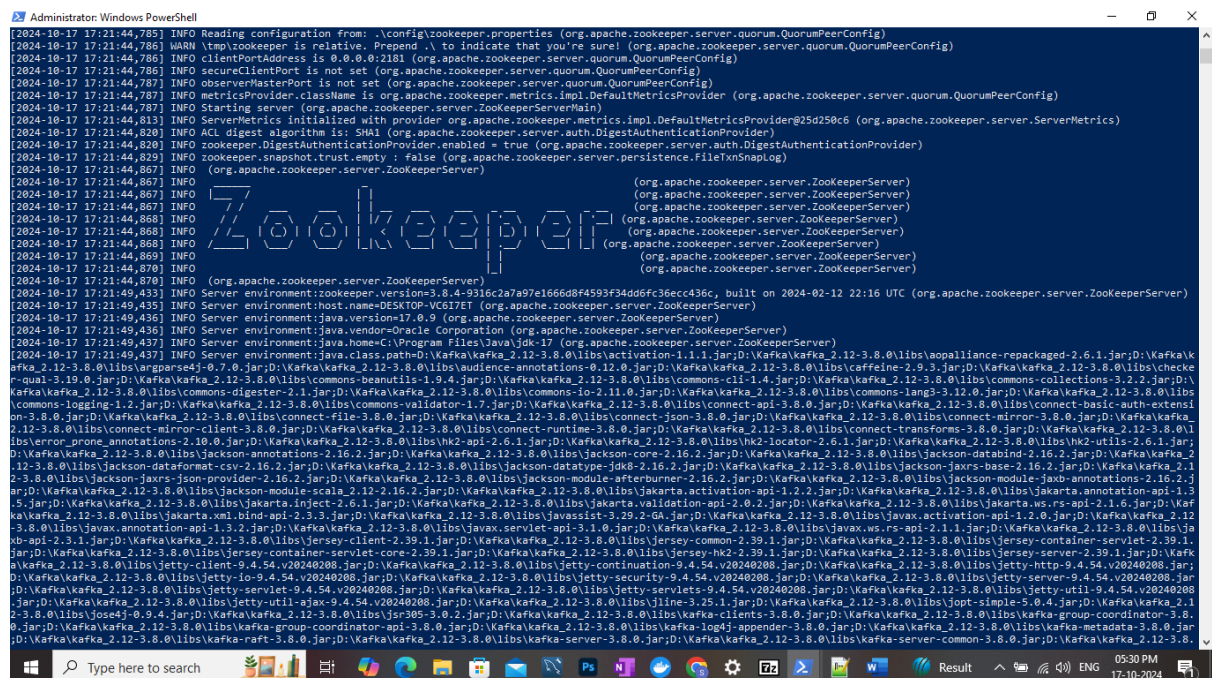
Then Open two windows Power shell as Admin and run below command

```
Administrator: Windows PowerShell

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

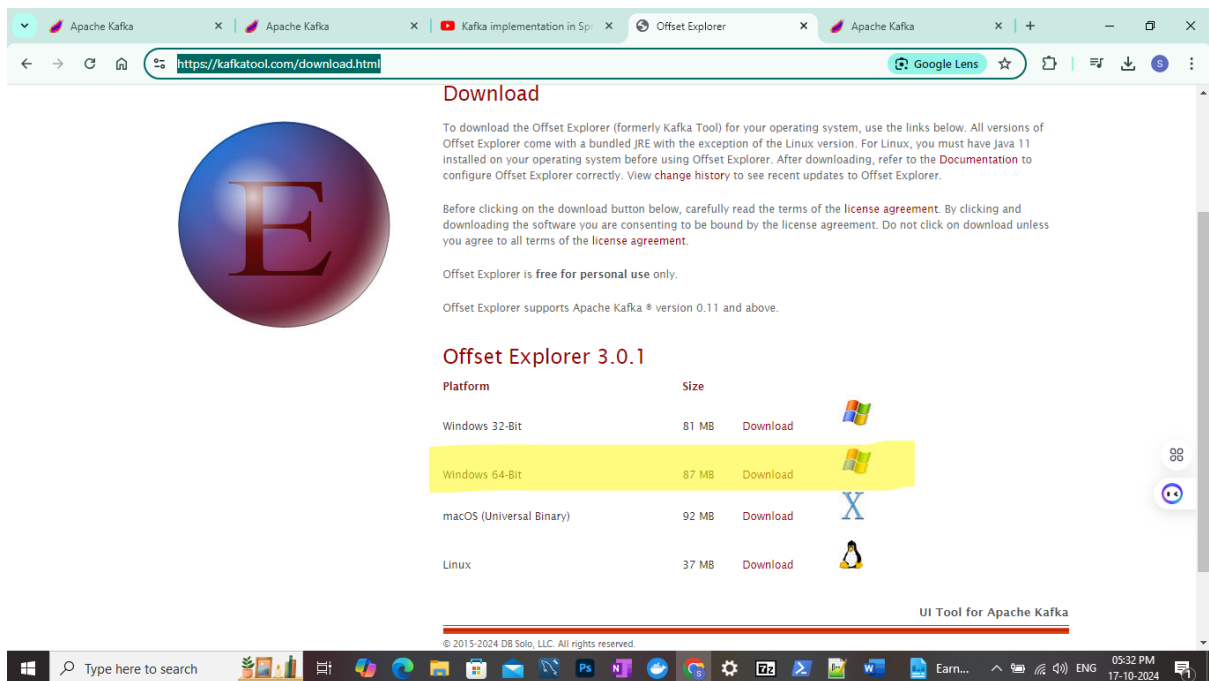
PS C:\Windows\system32> cd D:\Kafka\kafka_2.12-3.8.0
PS D:\Kafka\kafka_2.12-3.8.0> .\bin\windows\kafka-server-start.bat .\config\server.properties
```



```
cd D:\Kafka\kafka_2.12-3.8.0
.\bin\windows\zookeeper-server-start.bat .\config\zookeeper.properties (to run Zoo keeper)
.\bin\windows\kafka-server-start.bat .\config\server.properties (to run Server)
```

Go to below website

<https://kafkatool.com/download.html>



Download

To download the Offset Explorer (formerly Kafka Tool) for your operating system, use the links below. All versions of Offset Explorer come with a bundled JRE with the exception of the Linux version. For Linux, you must have Java 11 installed on your operating system before using Offset Explorer. After downloading, refer to the [Documentation](#) to configure Offset Explorer correctly. View [change history](#) to see recent updates to Offset Explorer.

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Offset Explorer is **free for personal use** only.

Offset Explorer supports Apache Kafka® version 0.11 and above.

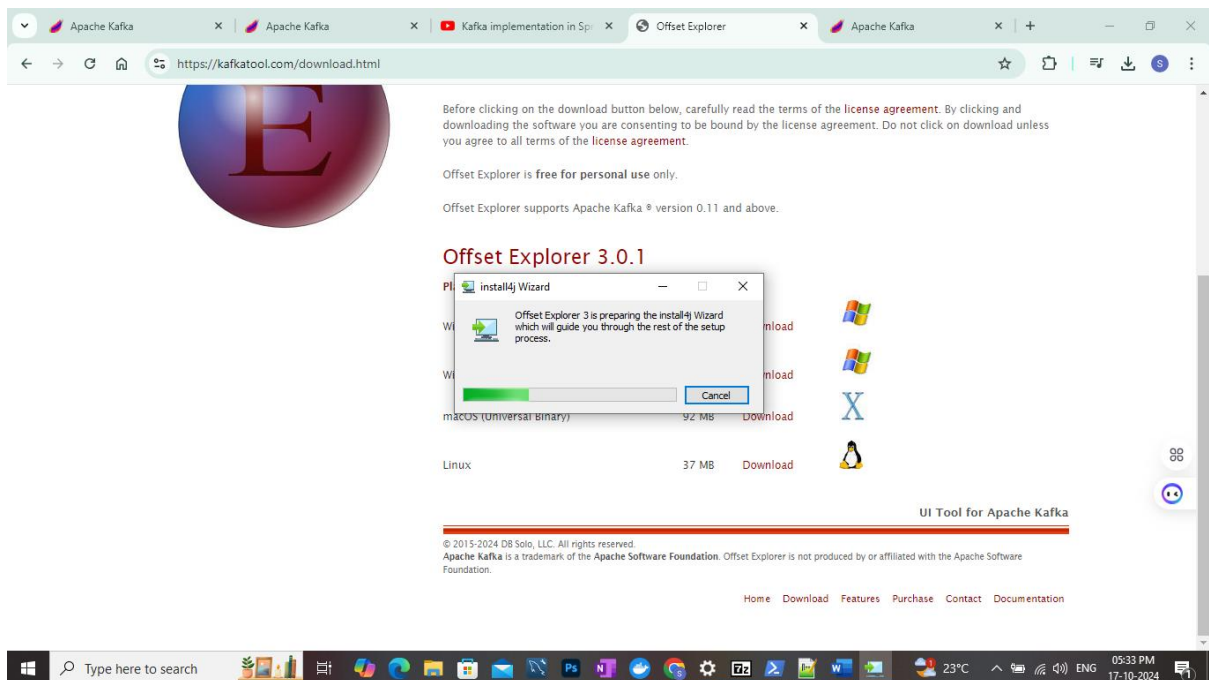
Offset Explorer 3.0.1

Platform	Size	Download
Windows 32-Bit	81 MB	Download
Windows 64-Bit	87 MB	Download
macOS (Universal Binary)	92 MB	Download
Linux	37 MB	Download

UI Tool for Apache Kafka

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Install it by clicking .exe file



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Offset Explorer 3.0.1

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Linux	37 MB	Download

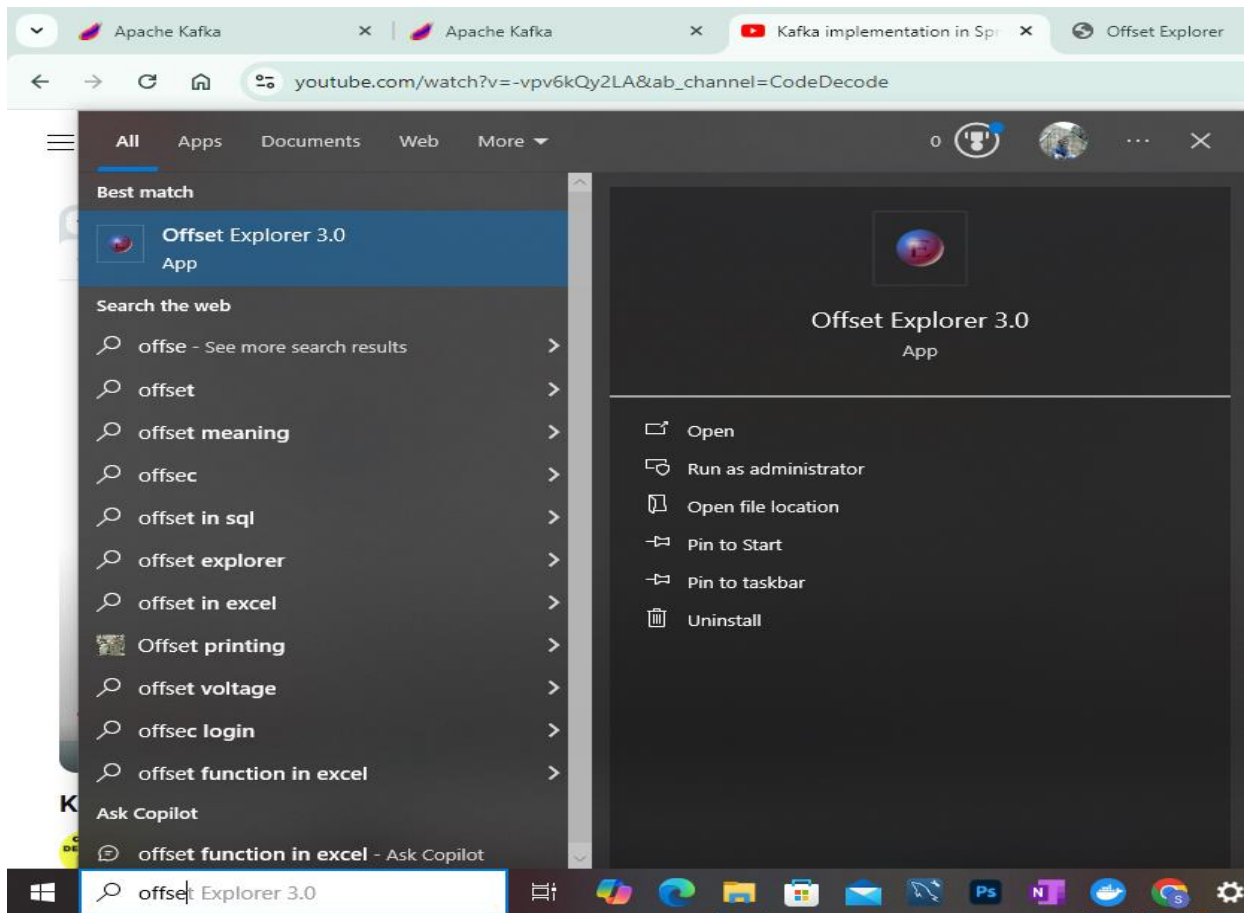
UI Tool for Apache Kafka

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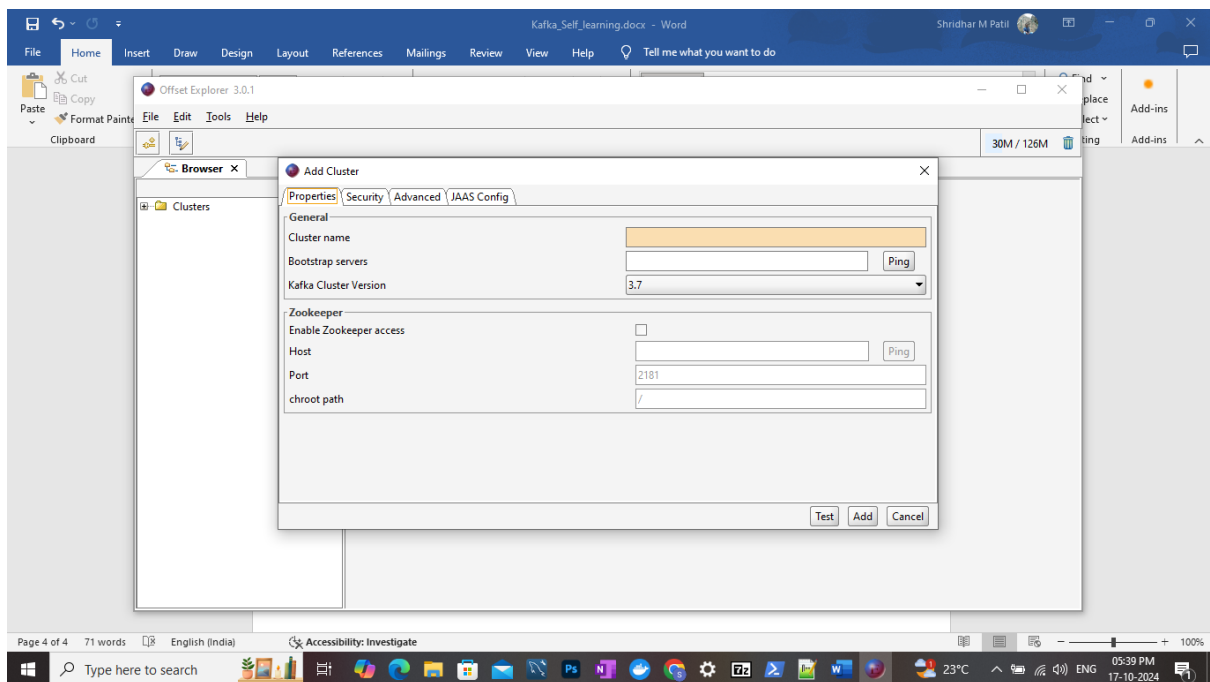
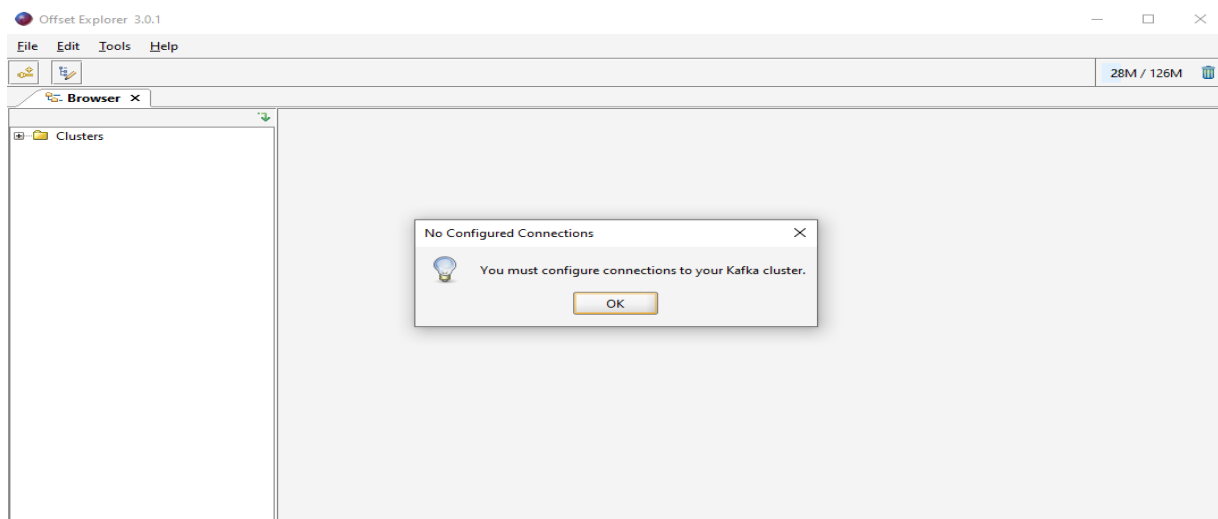
Apache Kafka is a trademark of the Apache Software Foundation. Offset Explorer is not produced by or affiliated with the Apache Software Foundation.

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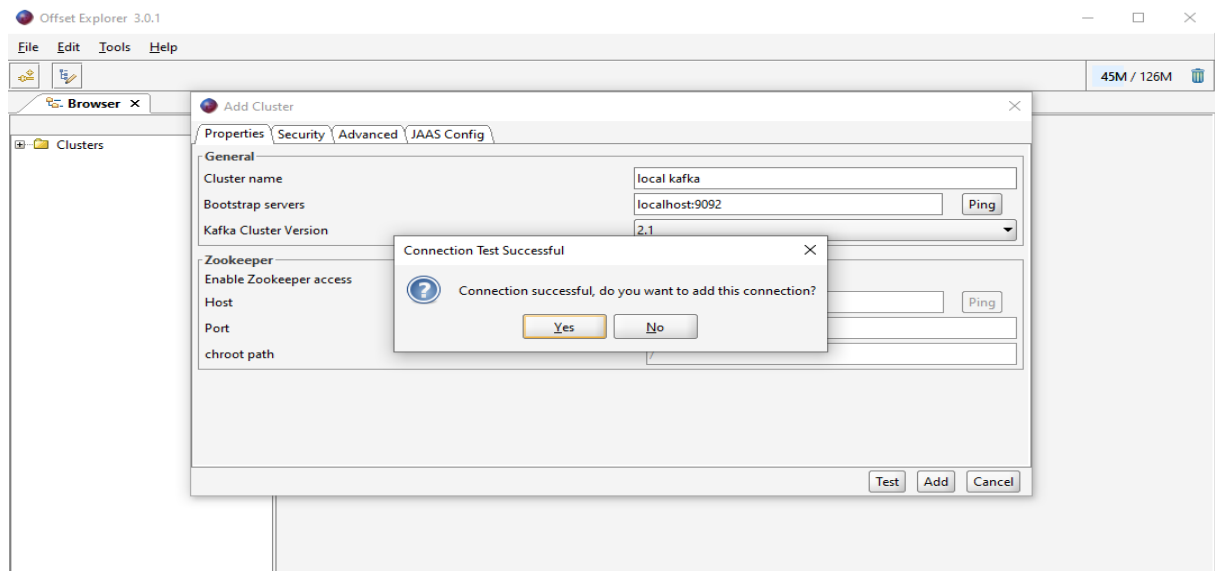
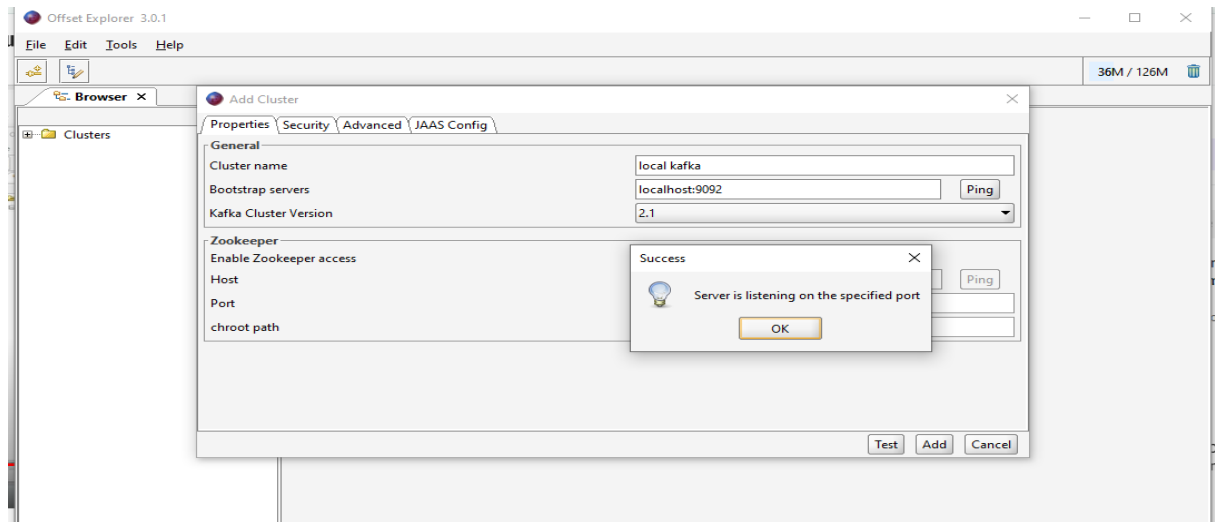
After installation search for offset explorer



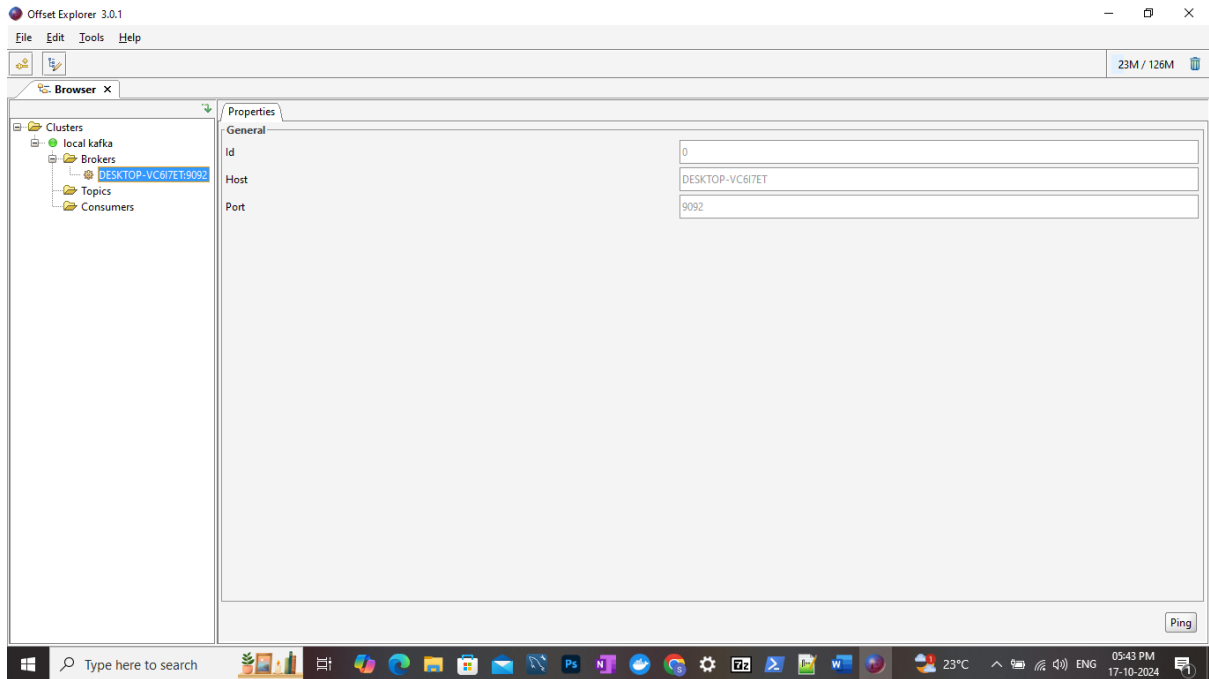
Open Offset Explorer



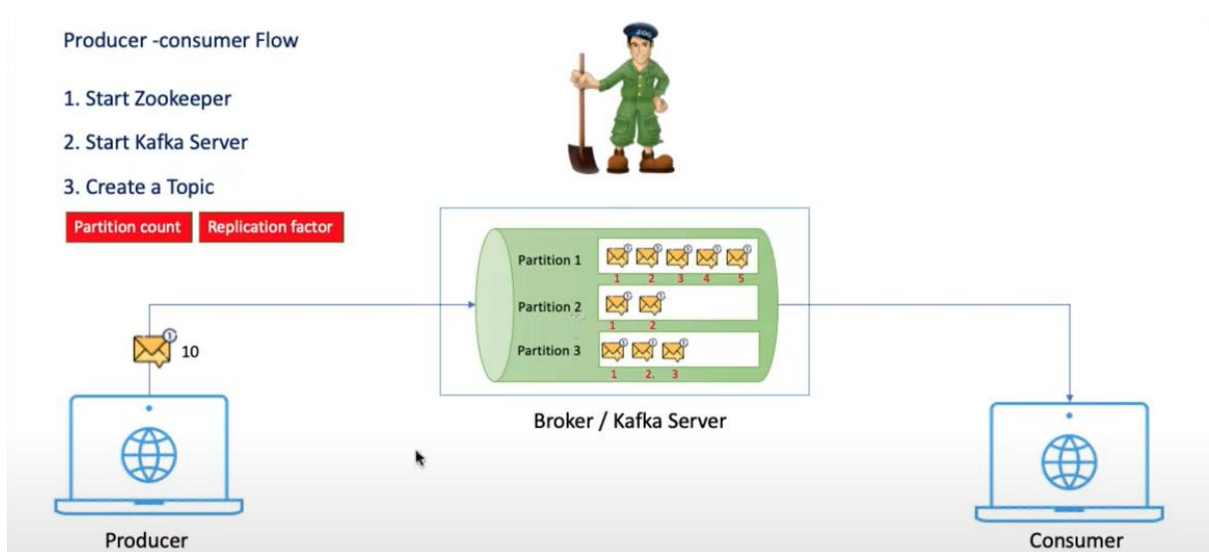
Add details and check ping



After Clicking on Test



Steps for Kafka to start



Kafka Demo.7z

How to start Zookeeper and kafka Server/Broker from command line prompt

```
cd D:\Kafka\kafka_2.12-3.8.0
.\bin\windows\zookeeper-server-start.bat .\config\zookeeper.properties (to run Zoo keeper)
```

```
. \bin\windows\kafka-server-start.bat . \config\server.properties (to run Server)
```

How to create topic and open kafka cli to check about topics/partition related details

```
. \bin\windows\kafka-topics.bat --bootstrap-server localhost:9092 --list
```

```
PS D:\Kafka\kafka_2.12-3.8.0> . \bin\windows\kafka-topics.bat --bootstrap-server localhost:9092 --list
MyKafkaTopic
Mytopic2
__consumer_offsets
```

```
. \bin\windows\kafka-topics.bat --bootstrap-server localhost:9092 --describe --topic topic-name
```

```
PS D:\Kafka\kafka_2.12-3.8.0> . \bin\windows\kafka-topics.bat --bootstrap-server localhost:9092 --describe --topic Mytopic2
[2024-10-20 12:24:42,015] WARN [AdminClient clientId=adminclient-1] The DescribeTopicPartitions API is not supported, using Metadata API to describe topics. (org.apache.kafka.clients.admin.Kaf
kaAdminClient)
Topic: Mytopic2 TopicId: cdDp3CBbRyCL7u-iSkj_gw PartitionCount: 3 ReplicationFactor: 1 Configs:
  Topic: Mytopic2 Partition: 0 Leader: 0 Replicas: 0 Isr: 0 Elr: N/A LastKnownElr: N/A
  Topic: Mytopic2 Partition: 1 Leader: 0 Replicas: 0 Isr: 0 Elr: N/A LastKnownElr: N/A
  Topic: Mytopic2 Partition: 2 Leader: 0 Replicas: 0 Isr: 0 Elr: N/A LastKnownElr: N/A
```

```
. \bin\windows\kafka-topics.bat --bootstrap-server localhost:9092 --create --topic Mytopic2 --
partitions 3 --replication-factor 1
```

Explanation:

--bootstrap-server localhost:9092: Connects to the Kafka broker running on localhost:9092.

--create: Creates a new topic.

--topic Mytopic2: The name of the topic being created (Mytopic2).

--partitions 3: Specifies that the topic will have 3 partitions.

--replication-factor 2: Sets the replication factor to 2, meaning each partition will be replicated across 2 brokers.

Open Producer and consumer from the console

Open Powe Shell or command prompt as administrator

```
cd D:\Kafka\kafka_2.12-3.8.0
```

```
//To Open Producer console
```

```
. \bin\windows\kafka-console-producer.bat --broker-list localhost:9092 --topic Mytopic2
```

```
//To run CSV file from CLI
```

```
. \bin\windows\kafka-console-producer.bat --broker-list localhost:9092 --topic Mytopic2 <D:\KafkaDemo
tips.csv (mac/linux)
```

```
Get-Content "D:/KafkaDemo/tips.csv" | .\bin\windows\kafka-console-producer.bat --broker-list localhost:9092 --topic Mytopic2 (Windows)
```

//To Open consumer console

```
.\bin\windows\kafka-console-consumer.bat --bootstrap-server localhost:9092 --topic Mytopic2 --from-beginning
```

```
cd D:\confluent-community-7.7.1\confluent-7.7.1 (TBD)
```

<https://github.com/Java-Techie-jt/kafka-installation/blob/main/README.md>

Using Kraft in Windows running Kafka without starting zookeeper server

```
$KAFKA_CLUSTER_ID = & ".\bin\windows\kafka-storage.bat" random-uuid
.\bin\windows\kafka-storage.bat format -t $KAFKA_CLUSTER_ID -c config/kraft/server.properties
.\bin\windows\kafka-server-start.bat config\kraft\server.properties
```

Command to decode the logs

```
.\bin\windows\kafka-dump-log.bat --cluster-metadata-decoder --files D:\tmp\kraft-combined-logs\_cluster_metadata-0\00000000000000000000.log
```

```
.\bin\windows\kafka-metadata-quorum.bat --bootstrap-server localhost:9092 describe --replication
```

 Administrator: Windows PowerShell

```
PS D:\Kafka\kafka_2.12-3.8.0> .\bin\windows\kafka-metadata-quorum.bat --bootstrap-server localhost:9092 describe --replication
NodeId  LogEndOffset  Lag  LastFetchTimestamp  LastCaughtUpTimestamp  Status
1       3350         0    1729567489271       1729567489271          Leader
PS D:\Kafka\kafka_2.12-3.8.0>
```

Using Docker image

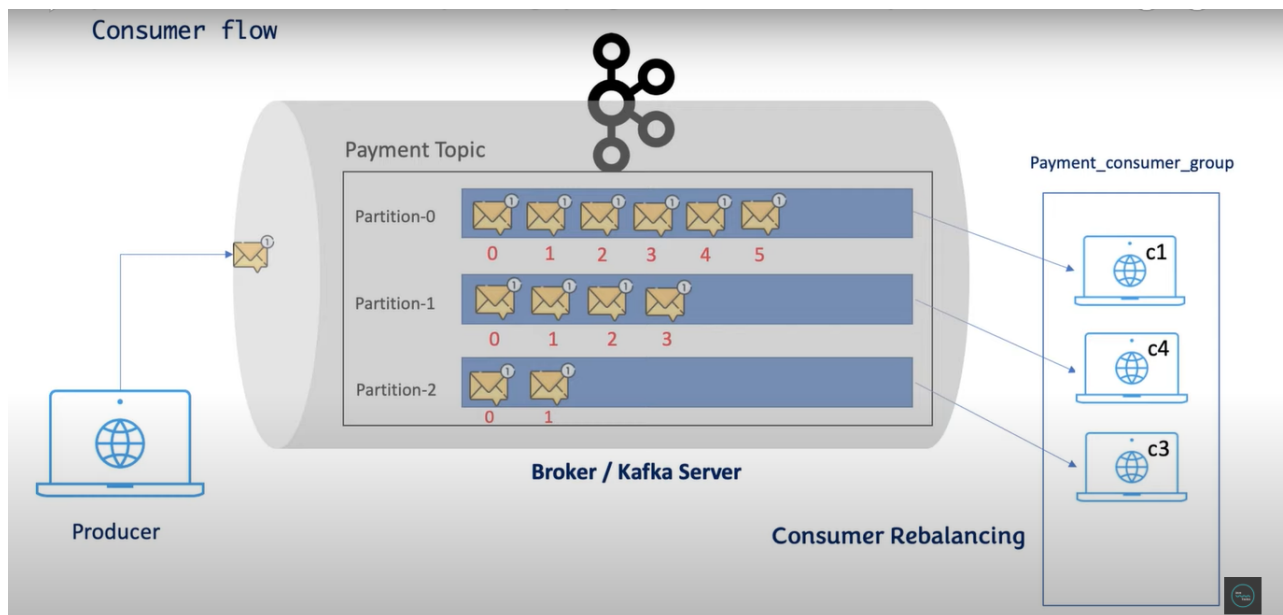
Docker-compose.yml

```
version : '3.1'

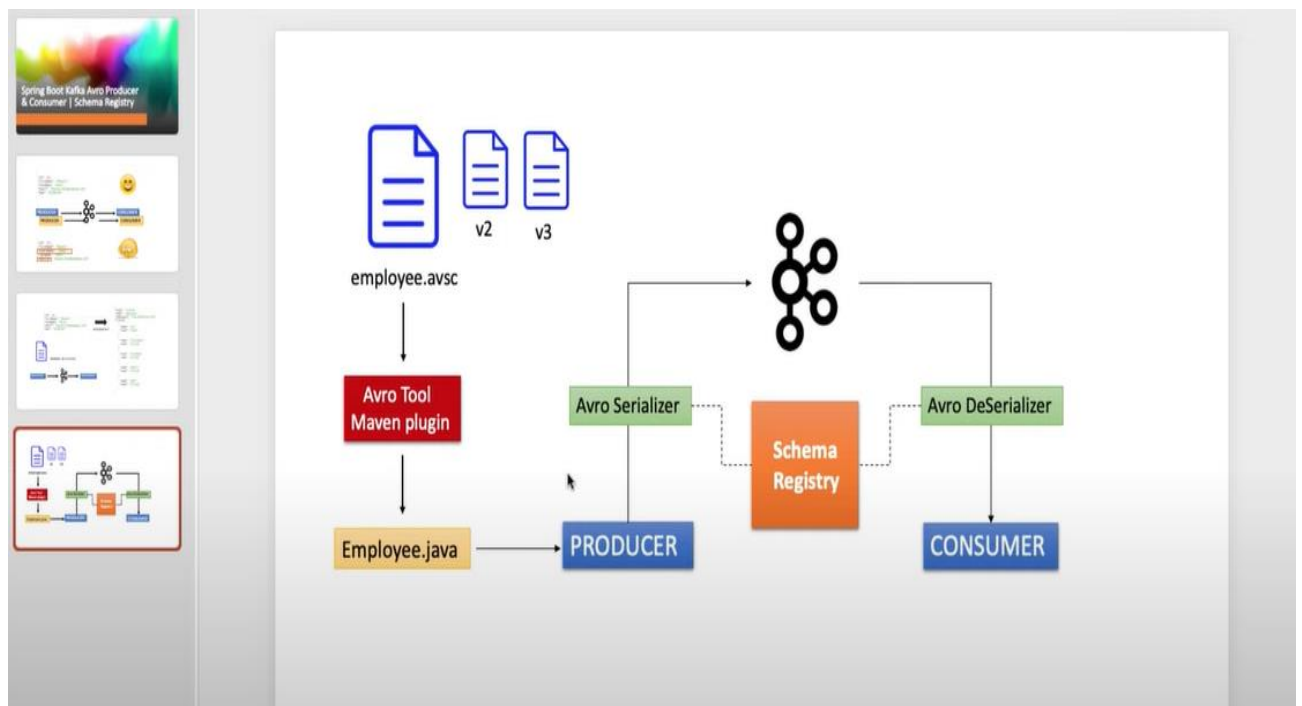
services:
  zookeeper:
    image: wurstmeister/zookeeper
    container_name: zookeeper
    ports:
      - "2181:2181"
  kafka:
    image: wurstmeister/kafka
    container_name: kafka
    ports:
      - "9092:9092"
    environment:
      KAFKA_ADVERTISED_HOST_NAME: localhost
      KAFKA_ZOOKEEPER_CONNECT: zookeeper:2181
```

Command to run docker-compose.yml

```
docker compose -f docker-compose.yml up -d
docker exec -it kafka /bin/sh
cd opt
cd kafka_2.13-2.8.1
cd bin
kafka-topics.sh --create --zookeeper zookeeper:2181 --replication-factor 1 --partitions 1 --topic quickstart
kafka-console-producer.sh --topic quickstart --bootstrap-server localhost:9092
kafka-console-consumer.sh --topic quickstart --from-beginning --bootstrap-server localhost:9092
```

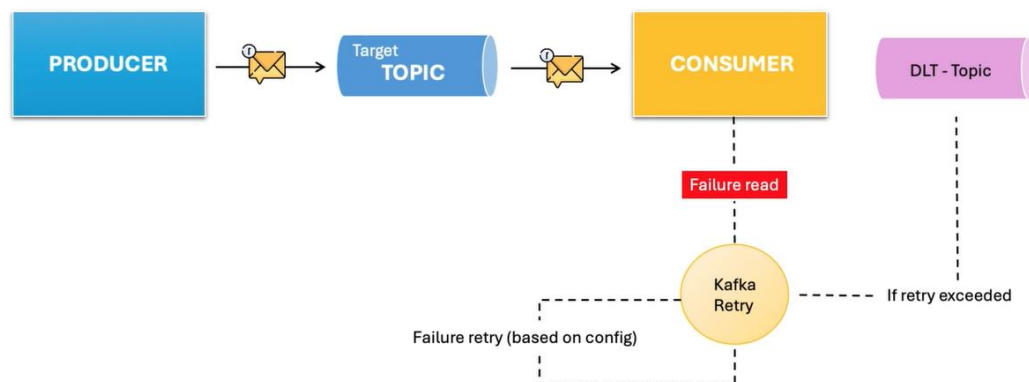



//Test Container to test Producer and Consumers are working properly



<https://github.com/Java-Techie-it/kafka-integration-test>

<https://github.com/Java-Techie-it/spring-kafka-avro>



<https://github.com/Java-Techie-it/kafka-error-handling>